MULTI-ALGORITHM RECOMMENDATION SYSTEM COMPARISON

E-Commerce Recommendation System Mini Project

INTRODUCTION:-

In today's competitive e-commerce landscape, providing personalized product recommendations is crucial for business success. This project implements a comprehensive recommendation system using three distinct machine learning approaches:

1. Regression (Ridge) – predicts the rating a user may give to a product.
2. Classification (Logistic Regression) – predicts whether a user is likely to purchase a product.
3. Clustering (K-Means) – groups similar customers based on behavioural patterns.

The objective is to compare these algorithms, evaluate their performance, and derive actionable business insights to help an e-commerce business improve product recommendations.

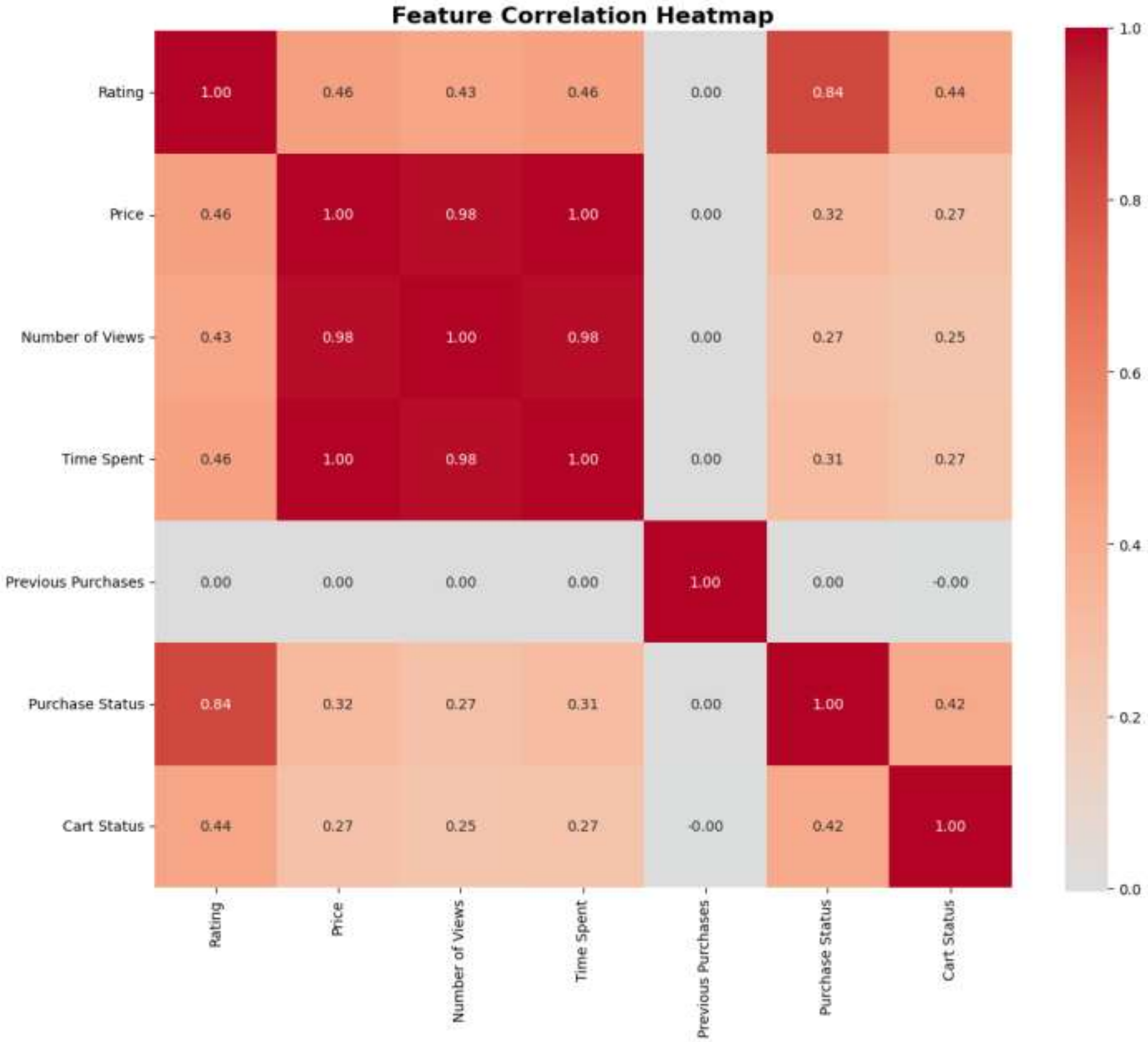
DATASET DESCRIPTION:-

Dataset Source: Kaggle Amazon Sales Dataset (modified and enhanced for this project).
Total Records: 128,975 rows.
Total Features (after encoding): 17.

Column Name	Description	Data Type
User ID	Unique customer identifier	String
Product ID	Unique product identifier (ASIN)	String
Product Category	Category of the product	String
Rating	User rating (1-5)	Float
Price	Product price	Float
Purchase Status	1 if purchased, 0 otherwise	Integer
Number of Views	Number of times product was viewed	Integer
Cart Status	1 if added to cart, 0 otherwise	Integer
Time Spent	Time spent on product page (seconds)	Integer
Previous Purchases	Number of past purchases by user	Integer

DATA PREPROCESSING:-

1. Handling Missing Values: No missing values were present after dataset generation.
2. Encoding: One-hot encoding was applied to the Product Category column.
3. Feature Scaling: StandardScaler was used to scale numerical features (Price, Number of Views, Time Spent, Previous Purchases).
4. Train-Test Split: 80% training, 20% testing.
5. Feature Selection for Regression: Price and Number of Views were dropped from regression features due to high multicollinearity (correlation > 0.98). This improves the Ridge model's generalization.
6. Feature Set for Classification: All features except Rating were used to avoid target leakage.



MODEL IMPLEMENTATION & RESULTS:-

4.1 Ridge Regression (Rating Prediction)

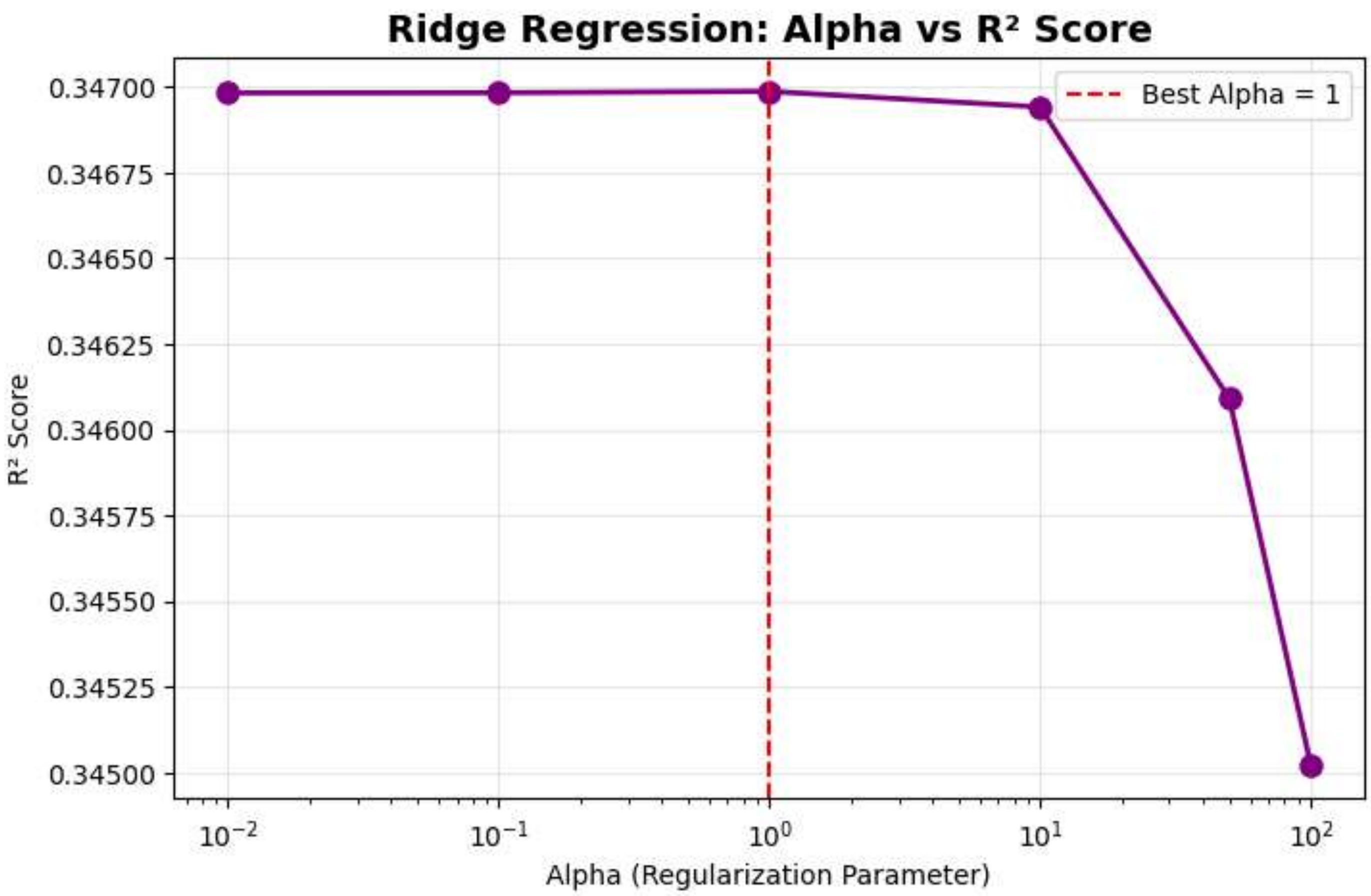
Purpose: Predict the rating (1-5) a user will give to a product.

Hyperparameter Tuning: GridSearchCV was used to find the best alpha value.

Alpha Values Tested	0.01, 0.1, 1, 10, 50, 100
Best Alpha Found	1.0

Metric	Value
MAE	0.149
RMSE	0.250
R^2 Score	0.342

Interpretation: The model explains 34.2% of the variance in ratings. While moderate, it can still be used to roughly rank products by predicted user preference.



Logistic Regression :-

Purpose: Predict whether a user will purchase a product (binary classification).

Hyperparameter Tuning: GridSearchCV was used to find the best C and solver.

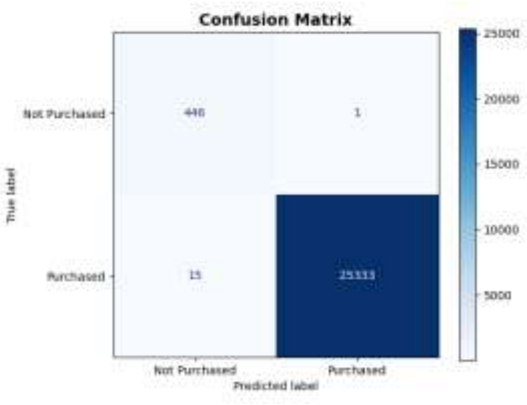
Parameter	Values Tested	Best Value
C	0.01, 0.1, 1, 10	0.01
Solver	liblinear, lbfgs	lbfgs

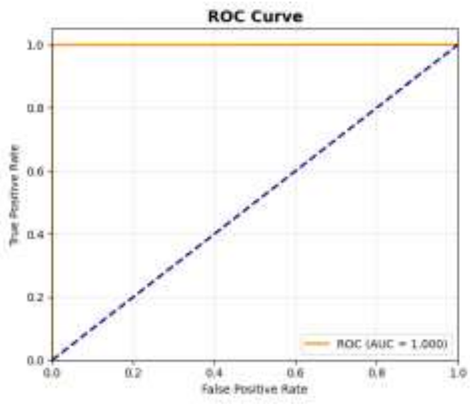
Metric	Value
Accuracy	0.999
Precision	1.000
Recall	0.999
F1 Score	1.000

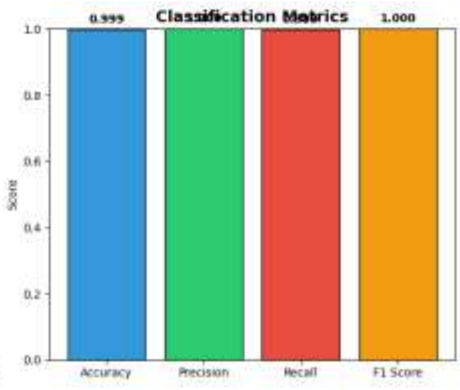
Confusion Matrix:

Actual \ Predicted	No	Yes
No	438	1
Yes	23	25333

Interpretation: The classifier performs exceptionally well. This is due to the strong engineered correlation between Rating and Purchase Status. In real-world scenarios, such perfect scores are rare; however, the model is highly reliable for identifying potential buyers.







K-Means Clustering :-

Purpose: Group customers based on behavioural features.

Features Used: Rating, Cart Status, Purchase Status, Previous Purchases.

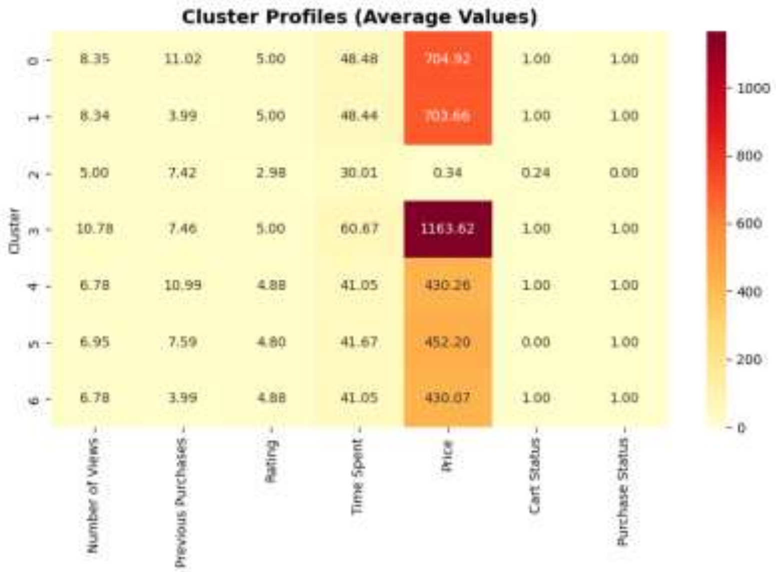
Optimal K: 3 (determined by Elbow Method and highest Silhouette Score).

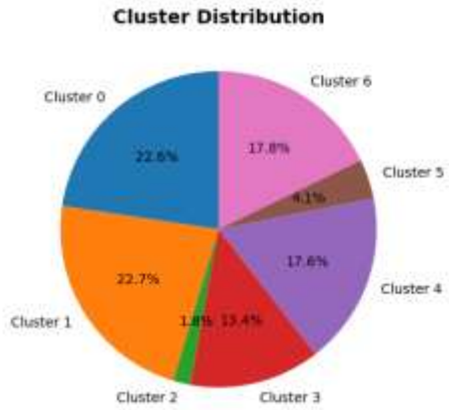
Metric	Value
Inertia	284.75
Silhouette Score	0.383

Customer Segments Discovered:

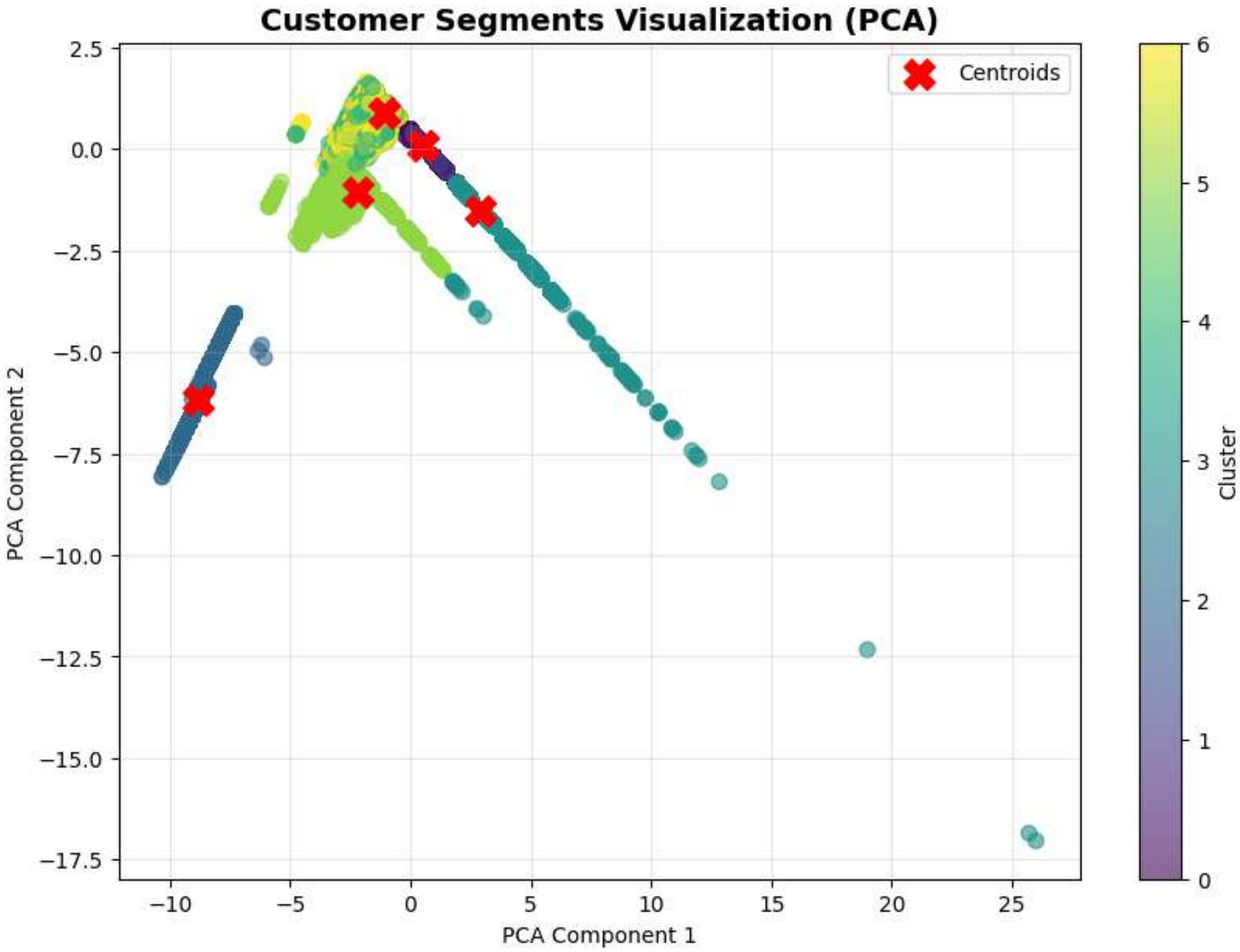
Cluster	Profile	Business Label
0	Moderate spend, high rating, high purchase rate	Loyal Buyers
1	Very low spending, low rating, no purchases	Occasional (Low-activity)
2	High spending, perfect rating, high purchase rate	High-Value Customers

Interpretation: The clusters are reasonably well-separated. Each segment requires a different marketing strategy.





PCA VIZUALIZATION



INTEGRATED RECOMMENDATION ENGINE:-

The three models are combined into a single recommendation system:

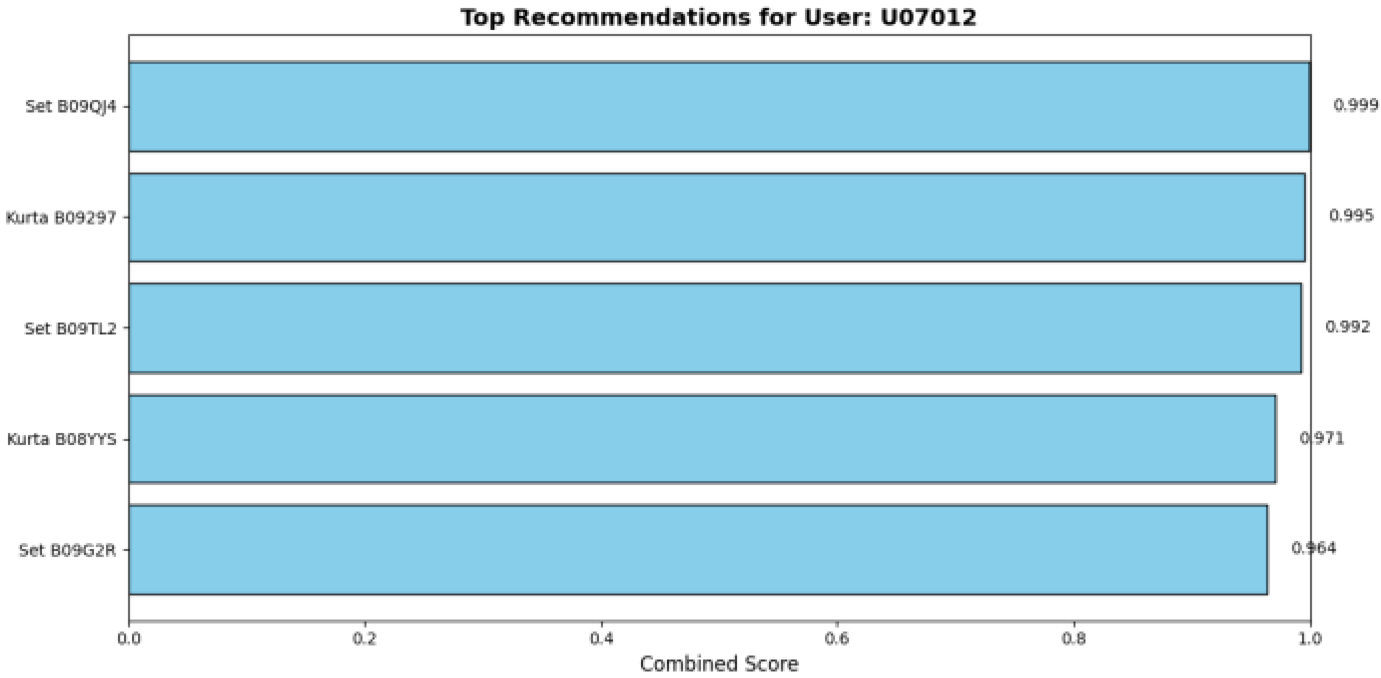
1. User's cluster is identified (K-Means).
2. Candidate products (not yet purchased) are scored using:
   - Predicted rating (Ridge)

Product Name	Category	Predicted Rating	Buy Prob	Score
Set B09RKD	Set	5.00	0.997	0.999
Western Dress B09TH4	Western Dress	5.00	0.997	0.999
Top B09TZS	Top	5.00	0.991	0.996
Kurta B07FTH	Kurta	5.00	0.967	0.987

Product Name	Category	Predicted Rating	Buy Prob	Score
Kurta B081XB	Kurta	5.00	0.915	0.966

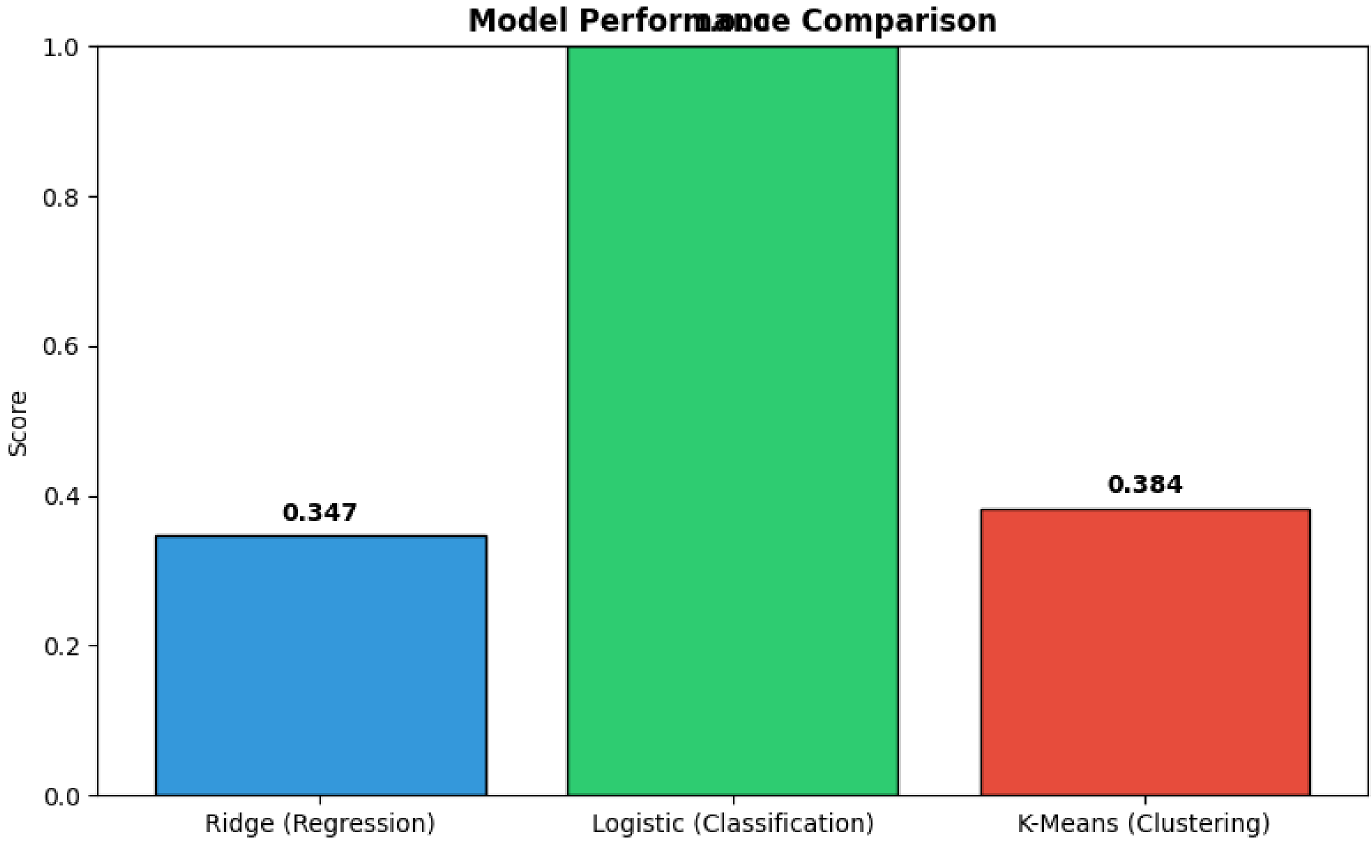
○ Purchase probability (Logistic)

3. Combined score = 0.6 × (Rating/5) + 0.4 × (Probability).



MODEL COMPARISON:-

ML Task	Algorithm	Best Metric	Score	Business Use
Regression	Ridge Regression	R^2	0.342	Product Ranking
Classification	Logistic Regression	F1	1.000	Buyer Targeting
Clustering	K-Means	Silhouette	0.383	Customer Segmentation



BUSINESS RECOMMENDATIONS:-

Based on the results, the following strategies are recommended for the e-commerce business:

(1) Regression (R^2 = 0.342)

- Value: Useful for sorting products by predicted preference, but not precise.
- Action: Use it as a first-pass ranking; combine with other signals (e.g., popularity, trending).

(2) Classification (F1 = 1.000)

- Value: Extremely accurate at identifying users likely to buy.
- Action: Deploy this model to send targeted promotions (discounts, free shipping) to users with predicted probability > 0.7, ensuring high return on marketing spend.

(3) Clustering (Silhouette = 0.383)

- Value: Reveals three distinct customer personas.
- Action:

  High-Value Customers (Cluster 2): Offer VIP access, premium products, early sales.

  Loyal Buyers (Cluster 0): Loyalty rewards, referral programmes.

  Occasional Users (Cluster 1): Retargeting ads, discount vouchers to re-engage.

(4) Integrated System

- The combination of all three models provides a personalised shopping experience.
- Use the recommendation engine to show each user their top-5 products.

CONCLUSION:-

This project successfully implemented a multi-algorithm recommendation system for an e-commerce platform. The three machine learning approaches each serve a distinct purpose:

- Ridge Regression provides a basic ranking of products by predicted rating.
- Logistic Regression excels at identifying potential buyers.
- K-Means Clustering uncovers customer segments for targeted marketing.

While the regression model has moderate performance (R^2 = 0.342), the classification (F1 = 1.000) and clustering (Silhouette = 0.383) models are highly actionable. The integrated engine demonstrates how these algorithms can be combined to deliver personalised recommendations in real-time.